

DynamicPed-VLN: A High-Fidelity Benchmark for Vision-and-Language Navigation Among Moving Pedestrians

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Abstract—Vision-and-Language Navigation (VLN) has made significant progress in static environments, but real-world applications require navigation in human-populated spaces with moving pedestrians. Recent benchmarks like HA-VLN 2.0 have pioneered human-aware navigation, yet challenges remain in achieving high-fidelity pedestrian dynamics and comprehensive language grounding. We present DynamicPed-VLN, the first high-fidelity VLN benchmark built on the community-standard Habitat 3.0 simulator with physics-consistent dynamic pedestrians. Our key innovations include: (i) a hybrid A*-ORCA trajectory planner that generates socially-compliant expert paths in crowded environments; (ii) a novel three-tier instruction generation system that provides hierarchical language supervision (Task-level Global Instructions, God-view Auxiliary Instructions, and Dual-layer Collaborative Instructions); and (iii) reinforcement learning and imitation learning baselines with explicit social cognition penalties (SCP) and pedestrian-aware fusion mechanisms. Comprehensive evaluation demonstrates that our benchmark presents realistic challenges for existing VLN methods, with state-of-the-art agents experiencing over 40% performance degradation in dynamic environments. Our framework achieves 55.1% success rate while maintaining 89.6% personal space compliance. The release of our benchmark, codebase, and pretrained models provides a foundation for developing socially-intelligent embodied AI systems that can safely operate in human-populated environments.

I. INTRODUCTION

Vision-and-Language Navigation (VLN) represents a fundamental capability for embodied artificial intelligence, enabling agents to interpret natural language instructions and execute goal-directed navigation in complex 3D environments [1, 2]. While significant progress has been made in controlled benchmarks, the transition from simulation to real-world deployment remains challenging due to the substantial domain gap between static environments and dynamic human-populated spaces [3, 4]. This gap is particularly pronounced in settings where robots must navigate while respecting social norms, adapting to unpredictable human behaviors, and maintaining situational awareness under partial observability [5, 6].

Despite the growing body of VLN research, current frameworks exhibit important limitations that impede real-world applicability. Traditional benchmarks predominantly assume static environments [7, 8], lacking dynamic human

interactions altogether. Recent pioneering work like HA-VLN 1.0 and HA-VLN 2.0 [9, 10] has made significant strides toward incorporating dynamic humans, establishing unified frameworks for discrete and continuous navigation.

However, even these advanced frameworks face critical challenges. First, their SMPL-based human animations—though improved—still suffer from mesh interpenetration and temporally discontinuous states in complex interactions or occlusion scenarios (e.g., humans walking behind walls or furniture), limiting visual and behavioral realism. Second, their expert trajectories are often generated without explicit social compliance mechanisms, leading to high collision rates (22.4% in dense scenes) and violations of personal space norms. Third, their instruction generation conflates activity semantics (e.g., “a person is walking”) with motion grounding, while also relying on unrealistic “God-view” priors that rarely occur in natural human–robot interaction—where commands are typically concise and goal-oriented (e.g., “go to the kitchen”).

Notably, these gaps persist despite recent advances in social navigation (SocialNav). Frameworks like Falcon [13] have established future-aware, physics-consistent robot–human co-navigation in point-goal tasks using Habitat 3.0, bidirectional social forces, and explicit Social Cognition Penalty (SCP). Yet Falcon—and most SocialNav works—operate without language grounding, leaving a critical question unanswered: Can socially intelligent mechanisms be effectively transferred to language-conditioned navigation?

Our work bridges this divide through DynamicPed-VLN, the first high-fidelity VLN benchmark built on the community-standard Habitat 3.0 simulator with a physically-realistic Spot robot model and dynamic, collision-avoidant pedestrians. Unlike prior VLN works that employ abstract cylindrical agents, we adopt Boston Dynamics’ Spot as the navigation platform—enabling realistic kinematics, sensor mounting (RGB-D at 0.83 m height), and physical interaction modeling (e.g., turning radius, acceleration limits). Humans are rendered via Habitat 3.0’s native HumanoidObject, supporting mesh-level collision, behavioral diversity (486 motion sequences), and active avoidance using ORCA—creating environments where agents must co-navigate with intelligent social actors, not scripted animations.

The significance of our contribution extends beyond technical novelty. By introducing pedestrian-aware language grounding, we redefine how instructions should be formulated in social navigation tasks. Rather than generic descriptions of human activities, our instructions explicitly

*This work was not supported by any organization

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reference and reason about specific pedestrians—mirroring how humans naturally communicate in crowded spaces. Our technical innovations include:

- 1) **Physically Grounded Simulation Platform:** We unify (i) a real-world robot model (Spot), (ii) high-fidelity human dynamics (Habitat 3.0 + ORCA), and (iii) ego-centric sensor suites (RGB-D, human count, BEV velocity)—establishing the first VLN benchmark where physical embodiment directly shapes perception, planning, and language grounding.
- 2) **Socially-Compliant Expert Trajectories:** A hybrid A*-ORCA planner generates reference paths that balance global efficiency with local social norms.
- 3) **Hierarchical Language Grounding:** Extending NavComposer [14], our speaker produces: (i) Task-level Global instructions (static, goal-oriented), (ii) Prior-Augmented instructions (motion-grounded, from short-term trajectory forecasting), and (iii) Dual-layer Collaborative instructions (global route + encounter-triggered hints), enabling ablation of linguistic support levels and mirroring real-world human-robot communication spectra.
- 4) **Socially-Aware Learning Baselines:** We provide IL (Dagger) and RL (DD-PPO + SCP) agents with explicit social compliance objectives, establishing strong baselines for future research.

Comprehensive evaluation reveals that state-of-the-art agents trained in static environments suffer $\sim 40\%$ performance degradation on our benchmark—highlighting the inadequacy of current evaluation paradigms. By releasing the dataset (28,000 instruction-trajectory pairs across 844 scenes), code, and models, we provide a robust foundation for developing socially intelligent embodied AI that can safely operate alongside humans in real-world settings.

II. RELATED WORK

A. Vision-and-Language Navigation in Continuous Environments

Early VLN benchmarks (e.g., R2R) rely on discrete navigation graphs, abstracting away low-level control and motion continuity. VLN-CE bridges this sim-to-real gap by requiring fine-grained continuous actions, yet assumes static, human-free environments—ignoring the core challenge of dynamic human co-navigation. Recent analyses confirm that the pedestrian dynamics gap, not visual fidelity, is the primary bottleneck for real-world transfer.

To improve robustness, hierarchical planning and modular grounding have emerged. IA-HWP introduces a decoupled waypoint planner (coarse view $\rightarrow$ refined location) with instruction-aligned supervision (RWO/DAL), mitigating shortest-path bias in long-horizon tasks. NavComposer further modularizes trajectories into action/scene/object entities and synthesizes instructions via LLMs—enabling controllable, diverse, annotation-free generation. Critically, both assume static scenes. While vision-language models like NaV-ILA extend egocentric reasoning to multi-image sequences,

they remain confined to human-free settings. Our work is the first to adapt these frameworks to dynamic pedestrian interaction, where waypoints must be socially certified—not just geometrically feasible.

B. Social Navigation Frameworks

Social navigation (SocialNav) traditionally focuses on point-goal tasks without language, emphasizing collision avoidance and social norm compliance. Falcon pioneers a future-aware framework in Habitat 3.0, introducing the Social Cognition Penalty (SCP)—including trajectory obstruction and proximity penalties—and auxiliary trajectory prediction tasks. It achieves 55

HA-VLN 2.0, the closest prior VLN benchmark with humans, uses a custom SMPL-based simulator lacking physics-consistent collision and bidirectional social forces. Its instructions emphasize activity semantics (e.g., “a person is walking”) rather than motion grounding, and provide no mechanism to ablate linguistic support levels. Our work bridges this divide: we are the first to transfer Falcon’s SCP and trajectory-aware planning to VLN, demonstrating that socially-intelligent mechanisms are not only compatible with—but essential for—language-guided navigation among moving pedestrians.

C. Instruction Generation and Interactive Augmentation

Instruction generation has evolved from end-to-end speakers to modular, entity-aware frameworks. NavComposer sets a new standard via entity decomposition (A/S/O), yet was designed for static environments. We extend it with a Pedestrian Entity Module, injecting motion-grounded priors (e.g., “approaching from left”, “slowing down”) via Qwen-VL—shifting from environment description to interaction description.

Beyond static generation, recent work explores adaptive instruction augmentation. StreamVLN (a.k.a. Help, Anna!) models human-in-the-loop assistance: when lost, the agent requests multimodal subtasks to resume navigation—mirroring our Level-3 Dual-layer Collaborative paradigm. Unlike StreamVLN’s reactive, on-demand help, our framework proactively structures instructions along three supervision tiers (Task-level Global / Prior-Augmented / Dual-layer Collaborative), enabling controlled ablation of linguistic support.

III. METHODOLOGY

A. High-Fidelity DynamicPed-VLN Dataset Generation Pipeline

We construct DynamicPed-VLN, the first vision-and-language navigation benchmark featuring physics-consistent dynamic pedestrians and motion-grounded hierarchical instructions, within the community-standard Habitat 3.0 simulator. The pipeline integrates three synergistic components:

1) *Expert Trajectory Generation: A*+ORCA Hybrid Planner*: Traditional VLN benchmarks use expert trajectories that ignore dynamic humans, limiting real-world applicability. Inspired by social navigation frameworks [13], we design a hybrid planner combining A*'s global optimality with ORCA's local social avoidance. Let the robot state at timestep t be $\mathbf{x}_t = (\mathbf{p}_t, \theta_t)$, goal location $\mathbf{g}$, and pedestrian set $\mathcal{H}_t = \{(\mathbf{h}_i, \mathbf{v}_i)\}$. **Global intent** is provided by the shortest-path waypoint $\mathbf{w}_k$ on the navigation mesh (computed via Habitat's built-in A* oracle), yielding the target heading adapted to Habitat's coordinate system:

$$\theta_{A^*} = \text{atan2}(w_k^z - p_t^z, w_k^x - p_t^x). \quad (1)$$

Local compliance is enforced by an ORCA-inspired velocity optimizer over *short-term predictions* of pedestrian states ($\tau = 2.0$ s ahead: $\tilde{\mathbf{h}}_i = \mathbf{h}_i + \tau \mathbf{v}_i$). The avoidance velocity $\mathbf{v}_{\text{avoid}}$ is computed as:

$$\mathbf{v}_{\text{avoid}} = \mathbf{v}_{\text{current}} + \sum_{i=1}^N \frac{\mathbf{v}_{\text{rep},i}}{N}, \quad (2)$$

where the repulsive velocity from pedestrian i is:

$$\mathbf{v}_{\text{rep},i} = \mathbf{v}_{\text{relative}} + \hat{\mathbf{r}}_{ij} \cdot \frac{(2r - \|\mathbf{r}_{ij}\|)}{\tau} \cdot \mathbb{I}[\|\mathbf{r}_{ij}\| < 2r], \quad (3)$$

with $\mathbf{r}_{ij} = \mathbf{p}_t - \mathbf{h}_i$, $r = 0.3$ m agent radius, and $\tau = 4.0$ s time horizon.

The final target heading is a *proximity- and speed-adaptive blend*:

$$\theta_{\text{tgt}} = \alpha_t \cdot \theta_{\text{avoid}} + (1 - \alpha_t) \cdot \theta_{A^*}, \quad (4)$$

where the fusion weight $\alpha_t \in [0.1, 0.95]$ uses piecewise-linear adaptation:

$$\alpha_t = \begin{cases} 0.95 & \text{if } d_{\min} < 0.3 \text{ m} \\ 0.8 + 0.1 \|\mathbf{v}_{\min}\| & \text{if } 0.3 \leq d_{\min} < 0.8 \text{ m} \\ 0.6 + 0.2 \|\mathbf{v}_{\min}\| & \text{if } 0.8 \leq d_{\min} < 1.5 \text{ m} \\ 0.3 + 0.1 \|\mathbf{v}_{\min}\| & \text{otherwise} \end{cases} \quad (5)$$

with $d_{\min} = \min_i \|\mathbf{p}_t - \mathbf{h}_i\|$ and $\mathbf{v}_{\min}$ its corresponding velocity.

Action mapping converts the target heading to discrete navigation actions based on angular thresholds:

$$\text{action} = \begin{cases} 1 \text{ (move forward)} & \text{if } |\theta_{\text{diff}}| < 15^\circ \\ 2 \text{ (turn left)} & \text{if } \theta_{\text{diff}} > 15^\circ \\ 3 \text{ (turn right)} & \text{otherwise} \end{cases} \quad (6)$$

where $\theta_{\text{diff}} = \theta_{\text{tgt}} - \theta_t$.

Safety enforcement applies three hard constraints:

- *Forward safety check*: Abort forward and output `stop` if predicted pose violates 0.8 m clearance with any pedestrian.
- *Anti-stuck recovery*: Force forward (if safe distance > 1.0 m) or lateral movement after ≥ 3 consecutive turns without progress; otherwise execute evasive turning.
- *Goal termination*: `stop` when $\|\mathbf{p}_t - \mathbf{g}\| \leq 1.0$ m, with automatic stop action appended to successful trajectories.

This planner reduces collision incidence from 22.4% (pure A*) to **1.8%** in scenes with avg. 6.8 pedestrians, while maintaining path efficiency (detour ratio: $1.12 \times$ shortest path). The implementation collects multi-sensor data including RGB, depth, third-person view, and human occupancy states for comprehensive VLN training.

Algorithm 1 Hybrid A*-ORCA Planner

```

1: procedure HYBRIDPLANNER( $\mathbf{x}_t, \mathbf{g}, \mathcal{H}_t$ )
2:    $\mathbf{w}_k \leftarrow \text{AStarOracle}(\mathbf{p}_t, \mathbf{g})$ 
3:    $\theta_{A^*} \leftarrow \text{atan2}(w_k^z - p_t^z, w_k^x - p_t^x)$ 
4:    $\mathbf{v}_{\text{avoid}} \leftarrow \mathbf{0}$ 
5:   for  $i \leftarrow 1$  to  $|\mathcal{H}_t|$  do
6:      $\tilde{\mathbf{h}}_i \leftarrow \mathbf{h}_i + \tau \mathbf{v}_i$ 
7:      $\mathbf{v}_{\text{rep},i} \leftarrow \text{ComputeRepulsiveVelocity}(\mathbf{p}_t, \tilde{\mathbf{h}}_i)$ 
8:      $\mathbf{v}_{\text{avoid}} \leftarrow \mathbf{v}_{\text{avoid}} + \mathbf{v}_{\text{rep},i} / |\mathcal{H}_t|$ 
9:   end for
10:   $\theta_{\text{avoid}} \leftarrow \text{atan2}(v_{\text{avoid},z}, v_{\text{avoid},x})$ 
11:   $\alpha_t \leftarrow \text{ComputeAdaptiveWeight}(d_{\min}, \|\mathbf{v}_{\min}\|)$ 
12:   $\theta_{\text{tgt}} \leftarrow \alpha_t \theta_{\text{avoid}} + (1 - \alpha_t) \theta_{A^*}$ 
13:   $\theta_{\text{diff}} \leftarrow \theta_{\text{tgt}} - \theta_t$ 
14:  if  $|\theta_{\text{diff}}| < 15^\circ$  and ForwardSafetyCheck() then
15:    return 1 ▷ Move forward
16:  else if  $\theta_{\text{diff}} > 15^\circ$  then
17:    return 2 ▷ Turn left
18:  else
19:    return 3 ▷ Turn right
20:  end if
21: end procedure

```

Algorithm 2 Large-Scale Expert Data Collection

Require: Config C , target episodes $M_{\max}$, max steps $T_{\max}$

Ensure: Dataset $\mathcal{D}$ with successful trajectories

```

1:  $\text{envs} \leftarrow \text{VectorEnvFactory}(C)$ 
2:  $\mathcal{D} \leftarrow \emptyset$ , collected  $\leftarrow 0$ 
3: Initialize buffers  $\text{buffers}[1..N] \leftarrow \emptyset$ 
4: while collected  $< M_{\max}$  do
5:    $\mathbf{O} \leftarrow \text{envs.reset}()$ 
6:    $\mathcal{E}_{\text{valid}} \leftarrow \text{FilterValidEpisodes}(\mathbf{O})$ 
7:   if  $\mathcal{E}_{\text{valid}} = \emptyset$  then
8:     continue
9:   end if
10:   $\mathbf{d} \leftarrow \text{False}^N$ ,  $\mathcal{E}_{\text{active}} \leftarrow \mathcal{E}_{\text{valid}}$ ,  $t \leftarrow 0$ 
11:  while  $\mathcal{E}_{\text{active}} \neq \emptyset$  and  $t < T_{\max}$  do
12:    for  $\text{env\_idx} \in \mathcal{E}_{\text{active}}$  do
13:       $a_t \leftarrow \text{HybridPlanner}(\mathbf{O}[\text{env\_idx}])$ 
14:      Store data in  $\text{buffers}[\text{env\_idx}]$ 
15:       $\text{envs.async\_step.at}(\text{env\_idx}, a_t)$ 
16:    end for
17:    for  $\text{env\_idx} \in \mathcal{E}_{\text{active}}$  do
18:       $\mathbf{o}, r, d, \mathbf{i} \leftarrow \text{envs.wait\_step.at}(\text{env\_idx})$ 
19:       $\mathbf{O}[\text{env\_idx}] \leftarrow \mathbf{o}$ ,  $\mathbf{d}[\text{env\_idx}] \leftarrow d$ 
20:      if  $d$  then  $\mathcal{E}_{\text{active}} \leftarrow \mathcal{E}_{\text{active}} \setminus \{\text{env\_idx}\}$ 
21:      end if
22:    end for
23:     $t \leftarrow t + 1$ 
24:  end while
25:  for  $\text{env\_idx} \in \mathcal{E}_{\text{valid}}$  do
26:    if SuccessCheck( $\text{buffers}[\text{env\_idx}]$ ) then
27:      AppendStopAction( $\text{buffers}[\text{env\_idx}]$ )
28:      SaveToDisk( $\text{buffers}[\text{env\_idx}]$ )
29:       $\mathcal{D} \leftarrow \mathcal{D} \cup \{\text{buffers}[\text{env\_idx}]\}$ 
30:      collected  $\leftarrow$  collected + 1
31:    end if
32:     $\text{buffers}[\text{env\_idx}] \leftarrow \emptyset$ 
33:  end for
34: end while
35:  $\text{envs.close}()$ 
36: return  $\mathcal{D}$ 

```

2) *Multi-Sensor Data Collection and Benchmark Construction*: Our data collection framework employs a sophisticated multi-sensor architecture designed for large-scale benchmark generation. The system leverages Habitat 3.0’s vectorized environment capabilities to enable efficient parallel data acquisition across multiple scenarios.

Sensor Suite Configuration: We instrument the robotic platform with a comprehensive multi-modal sensor suite, including an RGB camera (640×480 resolution) and a depth sensor for egocentric visual and geometric perception, a global third-person camera for trajectory visualization and behavioral analysis, dedicated pedestrian perception sensors—specifically a human-counting sensor and a velocity-estimation sensor—to support dynamic obstacle awareness, and a GPS-compass localization module to enable reliable point-goal navigation.

Quality Assurance Mechanisms: Algorithm 2 incorporates rigorous quality assurance protocols: trajectories with geodesic distances ≤ 3.0 m are automatically filtered out to ensure navigation complexity; successful episodes are validated only when the final robot-goal distance is ≤ 1.0 m; cross-sensor data integrity is verified through modality consistency and precise temporal alignment checks; and robust fault-tolerance mechanisms handle environment failures, simulation resets, and potential data corruption during large-scale parallel collection.

3) *Three-Tier Hierarchical Instruction Generation*: Building upon the NavComposer framework, We propose a comprehensive hierarchical instruction generation framework that leverages Qwen-VL’s multimodal capabilities to produce navigation guidance at three distinct supervision levels. The system processes RGB observation sequences $\mathcal{I} = \{I_1, I_2, \dots, I_N\}$ and corresponding action sequences $\mathcal{A} = \{a_1, a_2, \dots, a_T\}$ where $a_t \in \{\text{stop, move forward, turn left, turn right}\}$.

Intelligent Keyframe Selection: The framework employs a strategic keyframe selection algorithm to identify representative frames $\mathcal{K} \subset \{1, \dots, N\}$:

$$\mathcal{K} = \mathcal{K}_{\text{action}} \cup \mathcal{K}_{\text{pedestrian}} \cup \{1, N\} \quad (7)$$

where: $\mathcal{K}_{\text{action}} = \{t \mid a_t \neq a_{t-1}\}$ captures action transition points, $\mathcal{K}_{\text{pedestrian}} = \{t \mid \text{PedestrianDetected}(I_t) = \text{True}\}$ identifies frames with human presence, Sampling density adapts to episode length: $|\mathcal{K}| \in [2, 5]$ based on N .

Multi-Modal Analysis Pipeline: For each selected keyframe $k \in \mathcal{K}$, we perform comprehensive scene understanding:

$$\text{scene}_k = f_{\text{Qwen-VL}}(I_k, P_{\text{scene}}) \quad \text{Scene description} \quad (8)$$

$$\text{object}_k = f_{\text{Qwen-VL}}(I_k, P_{\text{object}}) \quad \text{Object/pedestrian localization} \quad (9)$$

$$\mathcal{M}_k = f_{\text{Qwen-VL}}(I_k, P_{\text{motion}}) \quad \text{Pedestrian motion analysis} \quad (10)$$

$$\mathcal{R}_k = f_{\text{Qwen-VL}}(I_k, P_{\text{risk}}) \quad \text{Collision risk assessment} \quad (11)$$

$$\mathcal{D}_k = f_{\text{Qwen-VL}}(I_k, P_{\text{density}}) \quad \text{Crowd density analysis} \quad (12)$$

where P_* represents specialized prompts for each analysis task.

Trajectory Prediction Module: For sequences of pedestrian frames $\mathcal{S} = \{(t_i, I_{t_i})\}$ with $|\mathcal{S}| \geq 2$, we perform multi-frame trajectory forecasting:

$$\mathcal{T}_{\text{pred}} = f_{\text{Qwen-VL}}(\{I_{t_i}\}_{i=1}^{|\mathcal{S}|}, P_{\text{trajectory}}) \quad (13)$$

predicting future pedestrian movements and optimal avoidance strategies.

Level 1: Global Instructions generate fundamental route descriptions using static environmental context combined with action sequences:

$$I_{\text{global}} = f_{\text{Qwen-VL}}(I_N, \mathcal{C}_{\text{static}}, \mathcal{A}_{\mathcal{K}}) \quad (14)$$

where: $\mathcal{C}_{\text{static}} = \bigcup_{k \in \mathcal{K}} \{\text{scene}_k, \text{object}_k\}$ aggregates scene and object descriptions across keyframes, $\mathcal{A}_{\mathcal{K}} = \{a_k \mid k \in \mathcal{K}\}$ represents the action sequence at selected keyframes.

This produces instructions like: *“Start from the living room, walk through the hallway past the staircase, turn left at the dining table, and stop in the doorway of the kitchen.”*

Level 2: Priori-Augmented Instructions incorporate dynamic pedestrian awareness and anticipatory guidance. The context integrates:

$$\mathcal{C}_{\text{dynamic}} = \bigcup_{k \in \mathcal{K}_{\text{pedestrian}}} \{\mathcal{M}_k, \mathcal{R}_k, \mathcal{D}_k\} \cup \mathcal{T}_{\text{pred}} \quad (15)$$

The instruction generation combines action sequences with multi-dimensional pedestrian analysis:

$$I_{\text{priori}} = f_{\text{Qwen-VL}}(I_N, \mathcal{A}, \mathcal{C}_{\text{static}}, \mathcal{C}_{\text{dynamic}}) \quad (16)$$

producing instructions that specify directional adjustments for proactive collision avoidance, such as: *“Turn left to avoid the pedestrian moving forward down the central aisle, then continue moving forward while adjusting slightly right to maintain clearance.”*

Level 3: Dual-layer Collaborative Instructions implement a real-time co-navigation paradigm where a virtual assistant provides context-aware guidance at pedestrian encounter moments, mimicking human collaborative navigation:

$$I_{\text{collab}} = I_{\text{global}} \oplus \bigoplus_{t \in \mathcal{T}_{\text{encounter}}} g_t(\mathcal{C}_{\text{dynamic}}, a_t, \mathcal{P}_t) \quad (17)$$

where: $\mathcal{T}_{\text{encounter}} = \{t \in \mathcal{K}_{\text{pedestrian}}\}$ denotes pedestrian encounter frames selected as keyframes, g_t generates real-time guidance mimicking a human co-navigator’s advice, $\mathcal{P}_t = \{\text{position, motion, trajectory}\}$ captures pedestrian state at encounter moment t

The collaborative guidance produces human-like instructions that reference specific pedestrians by observable characteristics (e.g., clothing, position, or motion), provide proactive directional adjustments based on predicted trajectories to ensure safe passage, maintain continuous navigation momentum by avoiding unnecessary stopping commands, and use natural, imperative language—as if spoken by a human

companion—to support intuitive and socially grounded co-navigation.

Example Collaborative Sequence:

- **Global:** "Navigate from the lobby to the conference room down the main corridor"
- **Encounter 1:** "As you approach the woman walking toward you from the left, steer slightly right to maintain clearance"
- **Encounter 2:** "The group ahead is moving slowly - pass them on the left side where there's more space"
- **Encounter 3:** "That person is about to cross your path - slow your pace and let them pass before continuing"

Algorithm 3 Hierarchical Instruction Generation Framework

```

1: procedure GENERATEINSTRUCTIONS(episode  $\mathcal{E}$ )
2:   Extract  $\mathcal{I}, \mathcal{A}$  from episode directory  $\mathcal{E}$ 
3:    $\mathcal{K} \leftarrow \text{SelectKeyframes}(\mathcal{I}, \mathcal{A})$   $\triangleright$  Action changes + pedestrian frames
4:    $\mathcal{C}_{\text{static}} \leftarrow \emptyset, \mathcal{C}_{\text{dynamic}} \leftarrow \emptyset$ 
5:   for each  $k \in \mathcal{K}$  do
6:      $\text{scene}_k \leftarrow \text{DescribeScene}(I_k)$ 
7:      $\text{object}_k \leftarrow \text{DescribeObject}(I_k)$ 
8:      $\mathcal{C}_{\text{static}} \leftarrow \mathcal{C}_{\text{static}} \cup \{\text{scene}_k, \text{object}_k\}$ 
9:     if PedestrianDetected( $\text{object}_k$ ) then
10:       $\mathcal{M}_k \leftarrow \text{AnalyzeMotion}(I_k)$ 
11:       $\mathcal{R}_k \leftarrow \text{AssessRisk}(I_k)$ 
12:       $\mathcal{D}_k \leftarrow \text{AnalyzeDensity}(I_k)$ 
13:       $\mathcal{C}_{\text{dynamic}} \leftarrow \mathcal{C}_{\text{dynamic}} \cup \{\mathcal{M}_k, \mathcal{R}_k, \mathcal{D}_k\}$ 
14:     end if
15:   end for
16:    $\mathcal{I}_{\text{pred}} \leftarrow \text{PredictTrajectories}(\mathcal{I}, \mathcal{K})$ 
17:    $\mathcal{C}_{\text{dynamic}} \leftarrow \mathcal{C}_{\text{dynamic}} \cup \mathcal{I}_{\text{pred}}$ 
18:    $I_1 \leftarrow \text{GenerateGlobal}(\mathcal{C}_{\text{static}}, \mathcal{A}, \mathcal{K})$ 
19:    $I_2 \leftarrow \text{GeneratePrior}(\mathcal{C}_{\text{static}}, \mathcal{C}_{\text{dynamic}}, \mathcal{A})$ 
20:    $I_3 \leftarrow \text{GenerateCollaborative}(I_1, \mathcal{C}_{\text{dynamic}}, \mathcal{A}, \mathcal{K})$ 
21:   return  $\{I_1, I_2, I_3\}$ 
22: end procedure
23: procedure GENERATECOLLABORATIVE( $I_{\text{global}}, \mathcal{C}_{\text{dynamic}}, \mathcal{A}, \mathcal{K}$ )
24:    $\mathcal{T}_{\text{encounter}} \leftarrow \{k \in \mathcal{K} \mid \text{PedestrianDetected}(I_k)\}$ 
25:    $\mathcal{G} \leftarrow \emptyset$   $\triangleright$  Collection of guidance snippets
26:   for each  $t \in \mathcal{T}_{\text{encounter}}$  do
27:      $\mathcal{P}_t \leftarrow \text{ExtractPedestrianContext}(\mathcal{C}_{\text{dynamic}}, t)$ 
28:      $a_t \leftarrow \mathcal{A}[t]$   $\triangleright$  Current robot action
29:      $h_t \leftarrow \text{GenerateEncounterGuidance}(\mathcal{P}_t, a_t, I_t)$ 
30:      $\mathcal{G} \leftarrow \mathcal{G} \cup \{(t, h_t)\}$ 
31:   end for
32:    $I_{\text{collab}} \leftarrow I_{\text{global}} \oplus \text{SortAndFormat}(\mathcal{G})$ 
33:   return  $I_{\text{collab}}$ 
34: end procedure

```

Encounter Guidance Generation: The guidance function employs Qwen-VL’s multimodal reasoning to generate human-like advice:

$$h_t = f_{\text{Qwen-VL}}(I_t, P_{\text{encounter}}, \mathcal{P}_t, a_t) \quad (18)$$

where $P_{\text{encounter}}$ is a specialized prompt template that elicits natural, imperative guidance mimicking human co-navigation behavior.

The three-tier framework enables systematic evaluation of navigation agents’ capabilities, from basic route following with action context (Level 1) to socially intelligent navigation with dynamic obstacle anticipation (Level 2) and real-time collaborative guidance (Level 3).

TABLE I
COMPARISON OF DYNAMICPED-VLN WITH EXISTING VLN BENCHMARKS. OUR BENCHMARK OFFERS SUPERIOR PEDESTRIAN FIDELITY, HIERARCHICAL INSTRUCTION STRUCTURE, AND SOCIAL COMPLIANCE METRICS WITHIN THE COMMUNITY-STANDARD HABITAT 3.0 PLATFORM.

Feature	R2R-CE	HA-VLN 2.0	Ours
Simulator	Matterport	Custom SMPL	Habitat 3.0
Pedestrian Fidelity	None	Medium	High
Avg. Pedestrians/Scene	0	4.2	6.8
Navigation Type	Continuous	Unified DE/CE	Continuous
Instruction Types	Single-layer	Activity-focused	Three-tier
Social Metrics	None	Basic	Comprehensive
Community Standard	✓	×	✓

B. Baseline Methods: RL and IL Frameworks

All baselines share a common architecture:

Visual Encoder: ResNet-50 processing RGB-D observations Language Encoder: CLIP text encoder for instruction embedding Pedestrian Perception: BEV occupancy map (10m×10m, 0.2m resolution) Fusion Network: LSTM module processing both global instructions and step-wise hints This architecture effectively integrates visual observations, language semantics, and pedestrian states to inform navigation decisions.

1) *IL-DAGger: Imitation Learning Baseline:* Our IL agent learns from A*+ORCA expert trajectories through Dataset Aggregation (DAGger) [16]:

Prioritizes sampling high pedestrian density scenarios Incorporates pedestrian awareness auxiliary tasks Balances expert demonstrations with student experience This approach achieves 58.3% success rate by closely following socially-compliant expert trajectories while generalizing to new environments.

2) *RL-SCP: Reinforcement Learning with Social Cognition:* Our RL agent employs DD-PPO [17] with a novel Social Cognition Penalty (SCP): Total Reward: This multi-component structure enables nuanced social behaviors beyond simple collision avoidance.

IV. EXPERIMENTS

A. Benchmark Construction and Statistics

DynamicPed-VLN contains 28,000 instruction-trajectory pairs across 844 scenes, with an average path length of 12.3m. Table 1 compares our benchmark with existing simulators, highlighting our advantages in environmental fidelity, human behavior modeling, and instruction quality.

B. Evaluation Metrics

We introduce two new metrics specific to dynamic pedestrian VLN: Pedestrian Reference Accuracy (PRA): Percentage of instructions where the agent correctly identifies and responds to referenced pedestrians

Dynamic Grounding Success (DGS): Percentage of episodes where the agent both reaches the goal and correctly handles pedestrian interactions mentioned in instructions.

These complement standard metrics (SR, SPL, TCR) to provide a complete evaluation of language-grounded social navigation.

C. Zero-Shot Transfer from Static to Dynamic Environments

Our most compelling result comes from evaluating state-of-the-art VLN agents trained in static environments on our dynamic benchmark.

D. Baseline Performance Analysis

E. Hierarchical Instructions Comparison

F. Ablation Studies

V. CONCLUSION

DynamicPed-VLN establishes a new paradigm for socially-aware vision-and-language navigation. By integrating high-fidelity human simulations, socially-compliant trajectory generation, and pedestrian-grounded language instructions within the community-standard Habitat 3.0 platform, we create the first benchmark that realistically captures navigation challenges in human-populated environments.

Our experiments reveal a stark reality: agents trained in static environments fail catastrophically when encountering dynamic pedestrians. This finding challenges the research community to reconsider evaluation methodologies and training paradigms for embodied AI.

The release of our benchmark, codebase, and pretrained models provides a foundation for developing socially-intelligent navigation systems that can safely operate alongside humans. Future work will extend our framework to outdoor environments, incorporate more complex human behaviors, and validate performance on physical robots through sim-to-real transfer.

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